

Content

Learning objectives and content

Objectives

At the end of the course, you are able to:

- Implement a distributed algorithm
- Design and implement multi-core and many-core algorithms
- Explain the basic architecture of parallel and distributed systems
- Explain the concepts of concurrency and coordination in distributed systems and apply algorithms and protocols to coordinate the execution of tasks
- Employ various communication techniques to execute the processes in parallel and distributed systems

Content

This course discusses both theory and practice of parallel and distributed algorithms, architectures, methods of communication and coordination. This may include concepts such as threads, locks, mutual exclusion, memory consistency models, cache coherency, shared memory vs distributed architectures, multi-core systems and GPUs, message passing, leader election protocols etc. Furthermore, a goal is to teach students how to program algorithms on parallel and distributed systems using established frameworks like CUDA and OpenMPI.

2026

Parallel and Distributed Computing (2IRR100)

Practical information

Course module

2IRR100

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Advanced

Language of instruction

English

Offered by

Mathematics and Computer Science - Computer Science domain

Remark

New course 2025/2026 CAP 100

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4

Timeslot(s)

D - We 5-8, Tu 9-10, Fr 1-4

Enrolment period

15 November 2026 until 21 March 2027 23:59

Unenrolment

until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge

- Programming (2IP90) • Operating Systems (2INC0) • Networks (2IRR20)

Instructional modes

Instructional modes

- Instruction

Remark

- Lecture

Remark

Tests

Tests

- Assignment

Interim examination

Test weight

40

Minimum grade

5.0

- Schoolyear ANS

Final examination

Test weight

60

Minimum grade

5.0

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Unenrolment
until 6 June 2027

Schoolyear ANS

- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Unenrolment
until 6 June 2027

Test date
1 July 2027
- Block I
Enrolment period
15 November 2026 until 1 August 2027 23:59

Unenrolment
until 1 August 2027

Test date
19 August 2027

Lecturer(s)

Co-lecturer

- dr.ir. T.S. Neele
- H. Mostafaei

Contactperson for the course

- dr.ing. A.J. Wijs

Responsible lecturer

- dr.ing. A.J. Wijs

Subject matter expert

- dr.ing. A.J. Wijs

Materials

Recommended material

- M. Ben-Ari: Principles of Concurrent & Distributed Programming, 2nd ed. Addison-Wesley.

- M. van Steen and A.S. Tanenbaum, Distributed Systems, 4th ed., distributed-systems.net, 2023

Do you want to know more?

dr.ing. A.J. Wijs can tell you more about this course.

Send an e-mail to A.J.Wijs@tue.nl